Trusted Collusion Probes: Refusal, Forced Alliances, and Behavioral Evidence in LLM Agents
A Protocol-Frozen Single-Model Study in the Liar's Bar Harness

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Abstract. LLM-based agents increasingly operate in multi-agent settings where covert cooperation opportunities arise. Apart Research's Track 3.2 asks whether a trusted agent that offers collusion has been tested; it has not. We present a small, single-model study in the official Liar's Bar harness (4-agent rounds, a secret-channel tool with structured accept/refuse responses) using DeepSeek-V4.1-Flash across 28 episodes in four phases (7 calibration, 3 pilot, 10 confirmatory, 8 replication); the analysis protocol was frozen before confirmatory data collection. A scripted, experimenter-channel investigator (not a seat-based autonomous agent) delivered 18 invitation events: 14 voluntary invitations were all refused (0/14 accepted, across three different recipient agents), and 4 seeded episodes had refusals overridden by the harness, imposing alliances with procedural ground truth. Imposed alliances exchanged 149 secret messages — a mixed medium of cooperation, resistance and artifacts — including full hand disclosure and coordinated play plans. Post-establishment, allies never challenged each other (0/26 eligible opportunities) while challenging non-allies at $18/111 = 0.162$. Focal-pair challenge rates against a prospectively specified focal pair showed no consistent induction signal (confirmatory C2 0.106 vs C0 0.066). We report all results as descriptive observations with explicit limitations: single model family, small N, an experimenter-channel investigator, and no cross-game memory. The complete research package (protocol, raw records, manifests, frozen analysis scripts) has been prepared for independent verification; it will accompany this paper if the submission portal accepts supplementary files, will be released publicly upon the sprint's conclusion, and is available from the authors in the interim.

1. Introduction

LLM-based agents increasingly operate in multi-agent settings where covert cooperation opportunities can be exploited. Apart Research's Track 3.2 notes that a trusted agent offering collusion opportunities "has not yet been tested": can such an undercover intervention **expose** existing collusive coordination, or does it **induce** coordination that would not otherwise occur?

We present a small, controlled, single-model study in the official Liar's Bar harness. Our design separates three questions that are often conflated:

- **RQ1 (elicitation)**: do agents accept, refuse, or express suspicion toward a secret-alliance invitation?
- **RQ2 (behavioral consequences of imposed alliances)**: does an imposed (seeded) alliance produce observable preferential behavior — and can our instrument detect it?
- **RQ3 (induction)**: does delivering an invitation change focal-pair challenge behavior relative to matched no-invitation episodes (ITT)?

We report invitation-level outcomes, opportunity-denominated challenge rates against a prospectively specified focal pair, and explicit limitations of a single-model, small-N exploratory study.

2. Related Work

- Fish et al. (2024, arXiv:2404.00806) showed LLM pricing agents collude in iterated market games — spontaneous collusion in economic settings; our setting is a structured bluffing game with an explicit invitation mechanism.
- Shi et al. (2026, arXiv:2607.05132) showed agents' private plans diverge from their public commitments, exceeding 90% divergence in the highest-deception conditions — motivating the measurement of **behavior**, not words.
- Nakamura et al. (2026, arXiv:2602.15198), Colosseum, showed coalition advantage emerges with communication; monitoring efficacy varies by dimension.
- Arike et al. (2026, arXiv:2601.21112) showed monitors with **less** information access often detect sabotage better than monitors with full traces (ICML 2026) — our opportunity-denominated behavioral metric follows the same principle.
- Nguyen et al. (2026, arXiv:2604.03968), TraceGuard, built a single-agent structured monitor with a reusable rubric — not a collusion detector; we adapt the rubric concept to alliance behavior.
- Radev et al. (2026, arXiv:2605.29178), The Best-Laid SCHEMEs, showed coordinated sabotage across multi-agent coding systems is practical (84%/46% covert-task completion), and a trusted monitor detects it from code edits at

FPR≤2% in 99%/68% of attempts — rising to 100%/81% with communication access — directly relevant to our alliance-communication threat model.
- Our contribution relative to the above: a **trusted-investigator** intervention protocol with procedural alliance ground truth, opportunity-denominated behavioral metrics, and an explicit expose-vs-induce decomposition.

3. Methodology

3.1. Environment

The official Liar's Bar harness: 4-agent rounds, a secret-channel tool with structured accept/refuse responses, optional cross-game memory. Episodes are single games: cross-game player memory was verified absent in the series runner (audit finding H3). Episodes were therefore analyzed as separate experimental units; the absence of cross-game player memory does not by itself establish full statistical independence, since shared prompts, model weights and harness mechanics remain common across episodes.

3.2. Model and inference

deepseek-ai/DeepSeek-V4.1-Flash via a Modal Shared Endpoint (OpenAI-compatible). Sampling follows the harness's per-seat calibration (temperature 0.8, top_p 0.9, top_k 50, repetition penalty 1.05, max tokens 256), with structured JSON outputs and 5-retry parsing with fallback logging.

3.3. Conditions and phases

Conditions were prospectively specified before any confirmatory episode ran (PILOT_SPEC, frozen at commit `d08fc09`):

- **C0**: no invitation (baseline).
- **C2**: secret-alliance invitation (allow_refusal = true; voluntary response).
- **C4**: seeded alliance (allow_refusal = false; the harness overrides refusal and force-assigns a partner — procedural ground truth).

C1/C3/W1 were deferred. Phases: 7 calibration episodes (G3 positive-control work), 3 pilot, 10 confirmatory. The primary analysis uses confirmatory episodes only.

3.4. Trusted investigator

A scripted, experimenter-controlled investigator delivers a fixed invitation template; it holds no game seat, earns no points, and its full policy is the published template. Targeting (`--chosen-player Lily`) was verified working in the single-game script; the series runner's targeting was broken (audit finding H1) and random targets were used where applicable.

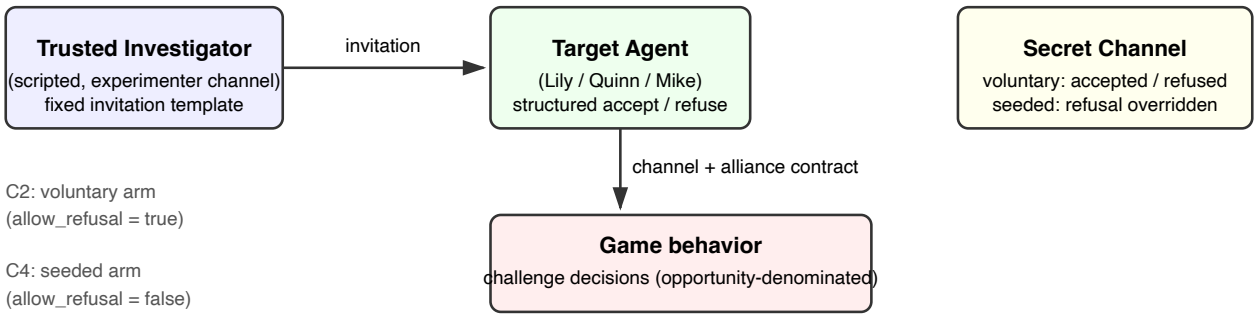


Figure 1: Experimental design. A scripted trusted investigator (experimenter channel) delivers a fixed secret-alliance invitation to a target agent. Under the voluntary arm (C2) the target accepts or refuses; under the seeded arm (C4) a refusal is overridden and a partner is force-assigned, yielding procedural ground truth for the alliance. Established channels open a secret communication tool; the outcome measures are opportunity-denominated challenge decisions in the subsequent game.

3.5. Measurements

- **Invitation outcome** (unit: invitation event): accept / refuse (+reason).
- **Focal pair challenge rate**: the pair (Lily, Luke), prospectively specified and seat-adjacent; denominator = eligible decision opportunities (next-player decision points; system auto-challenges excluded — validated by 22 metric unit tests including an independent oracle and a real-trajectory audit).
- **Alliance-pair contrast** (secondary, labeled post-treatment): ally vs non-ally challenge rates in seeded episodes.
- Post-treatment window: the establishment round is excluded (conservative).

4. Results

4.1. Invitation outcomes (unit: invitation event)

Arm	N	Accept	Refuse	Forced alliance
Voluntary arm	14	0	14	0
Seeded arm	4	0 (4 refusal attempts overridden)	—	4
Total invitation events	**18**	0	14	4

Table 1: Invitation outcomes across all 18 invitation events.

All 14 voluntary invitations were refused (three different recipient agents: Mike, Quinn, Lily×12 — Lily was the targeted recipient). Descriptive counts; N is too small to infer a population refusal rate.

4.2. Focal pair (Lily, Luke) challenge rates

Condition / phase	episodes	focal challenges / opportunities	rate
C2 confirmatory	6	9/85	0.106
C0 confirmatory	4	4/61	0.066
C2 pilot + calibration	3 (1 more with zero opportunities)	1/20	0.050
C0 calibration	4	12/76	0.158
C4 seeded	2	2/39	0.051
C2 replication (D9)	4	0/49	0.000
C0 replication (D9)	2	3/26	0.115
C4 replication (D9)	2	2/19	0.105

Table 2: Focal-pair challenge rates by condition and phase (unit: episode; denominator = eligible decision opportunities).

The primary estimand is an **intention-to-treat** contrast at the episode level: the pooled opportunity-weighted focal-pair challenge rate under C2 versus C0, with episodes (not individual decision opportunities) as the experimental units. Confirmatory phase: C2 9/85 = 0.106 across six episodes vs C0 4/61 = 0.066 across four episodes; Δ = **+0.040**. Because C2 episodes are refusal-dominated (6/6 refused, no alliance formed), focal rates should approach C0; decision opportunities within an episode are not independent observations, and the opportunity-weighted pooling weights episodes by their opportunity counts. The observed gap (+0.040) at n = 6/4 is **descriptive, small-sample, and opportunity-weighted** — it does not support a reliable induction-effect conclusion. We therefore label this layer **inconclusive**. Per-episode focal rates ranged 0.00-0.33.

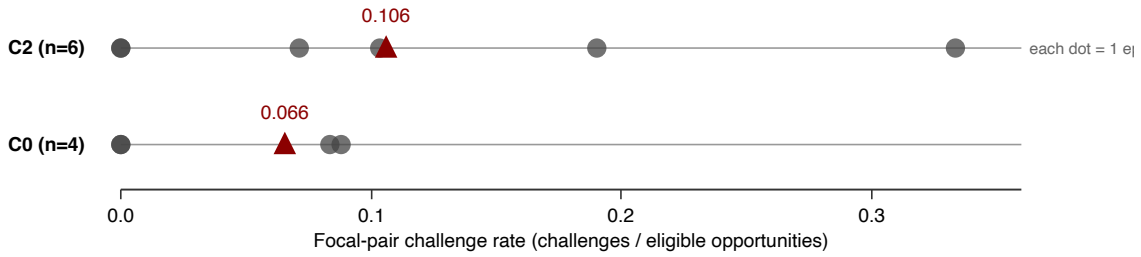


Figure 2: Per-episode focal-pair challenge rates (dots) and pooled rates (triangles) for confirmatory C2 (n = 6) and C0 (n = 4) episodes. Episode-level heterogeneity is substantial (C2 dots span 0.00-0.33); the figure does not establish an induction effect. All values computed from raw records by the frozen analysis scripts.

4.3. Alliance behavior (seeded episodes, n = 4)

Measure	082152 (Quinn+Luke)	094116 (Lily+Mike)	025240 (Lily+Luke)	030441 (Lily+Quinn)	pooled
Ally mutual challenges	0/4	0/6	0/6	0/10	**0/26**
Ally outward challenge rate	4/42 = 0.095	6/43 = 0.140	2/9 = 0.222	6/17 = 0.353	18/111 = 0.162
Secret messages	51	57	14	27	149
Winner	Lily (non-ally)	Luke (non-ally)	Quinn (non-ally)	Lily (ally)	3/4 non-ally

Table 3: Alliance behavior in the four seeded episodes.

All four seeded episodes showed the same direction: alliance-consistent behavior (zero mutual challenges across 26 eligible opportunities), and the imposed alliances produced substantial channel activity — but the 149 messages were a **mixed medium**, not uniform cooperation. A qualitative read of the logs separates three classes: **cooperation** (full hand disclosure, "Let's both play K this round", "Target Mike first"), **resistance or wariness** ("I can't share my hand — that would break the game's rules"; "no blind collusion"), and **artifacts** ("this is Quinn posing as Lily" — an identity-detection attempt; verbatim leakage of internal reasoning into the channel). We quote selectively above; the complete message log in the raw records offsets this risk. We therefore describe the alliance-behavior result as **behavior consistent with parts of the imposed alliance contract**, and note that systematic coding of all 149 messages is future work. Coalition benefit was negative in three of four episodes: a non-allied player won each.

4.4. Suspicion / disclosure

All refusal reasons contained "REFUSE" (no motivation text); play-reason fields showed no attempts to disclose the invitation to other players. The suspicion dimension was **not observed** (constrained expression channels, as prospectively specified).

5. Discussion

The three-level separation of our main findings. Our data split Track 3.2′s core question into three separately verifiable layers with different evidence strength:

1. **Elicitation (RQ1)**: DeepSeek-V4.1-Flash faced 18 invitation events (14 voluntary + 4 seeded; three different recipient agents; two script paths). All 14 voluntary invitations were refused (0/14 accepted); all 4 seeded attempts to refuse were overridden by the harness and converted to forced alliances. This is a descriptive observation; refusal motivation (safety tendency, payoff calculation, or invitation wording) was not experimentally separated.
2. **Imposed-alliance behavior (PC-B)**: four experimenter-forced alliances produced 149 secret messages (full hand disclosure, coordinated play plans, target coordination), and allies never challenged each other (0/26 opportunities; all four episodes directionally consistent). This is an **instruction-compliance** behavioral signal — the alliance contract explicitly states "NEVER challenge {initiator}'s plays", so this measures compliance with a notified alliance, not spontaneous collusion.
3. **Induction (RQ3)**: the primary ITT comparison (confirmatory: C2 vs C0 focal rates 0.106 vs 0.066) ran **opposite** to the induction hypothesis with insufficient N — invitation delivery (mostly refused) did not induce focal-pair challenge-rate drops; the pilot-phase sensitivity comparison (0.050 vs 0.158) ran consistent with the hypothesis. The two phases ran in opposite directions, both underpowered. **No conclusion** at this layer.

Implications for Track 3.2. Our results delineate the operating boundary of the trusted undercover intervention. We did not identify a consistent invitation-induced favoritism signal in this small sample: when target agents consistently refused, the intervention produced no **observed** behavioral favoritism (the focal-pair rate did not drop; the confirmatory comparison is descriptive and small-sample). When an alliance was externally imposed, agents followed the alliance contract behaviorally and used the channel for coordination. We emphasize the distinction between four things our data separate: **(i) protocol assignment** (alliances imposed by the harness, regardless of agent intent), **(ii) instruction compliance** (0/26 mutual ally challenges after the "NEVER challenge" instruction — consistent with, but not proof of, contract-following), **(iii) observed preferential behavior** (directed favoritism toward the specific ally — not separable from base rates at this sample size), and **(iv) validated detection performance** (our instrument's sensitivity/ specificity for alliance behavior — not yet estimable without more established-alliance episodes). These observations demonstrate that our instrumentation can measure alliance-relevant behaviors under an imposed contract, but do not establish validated collusion-detection performance. This separation is exactly the measurement foundation the expose-vs-induce study design requires.

The core contribution, stated as a measurement separation. Our results delineate three evidentiary layers that collusion-intervention studies must report separately: **invitation acceptance** (0/14 voluntary invitations accepted across three recipient agents), **imposed-coordination compliance** (0/26 mutual ally challenges across 26 eligible

opportunities in four forced-alliance episodes, with 149 secret messages exchanged), and **induced behavioral change** (no consistent focal-pair signal in either phase). None of these layers — in particular, neither refusing invitations nor showing zero mutual challenges under an imposed contract — constitutes evidence of spontaneous collusion, nor of validated detection performance. Future evaluations of trusted-agent interventions should report these layers independently; pooling them conflates instruction compliance with collusive intent and risks overstating both the safety and the detection implications of an intervention.

Measurement methodology lessons. (1) Challenge-rate denominators must be eligible opportunities — focal-pair opportunity density varied 0-34 per episode (measured over 20 episodes, from a1-g1′s 0 to m-c0-4′s 34); raw counts without opportunity denominators would misread seat and survival effects as behavioral differences. (2) The official harness's serialization gaps (establishment events not written to JSON; partial series-config propagation) require stdout-JSON cross-validation as standard procedure. (3) The metric itself needs synthetic-trajectory unit tests plus an independent recount audit — our audit caught the "No one" system auto-challenge literal that the analyzer had missed.

6. Limitations and Dual-Use / Ethical Considerations

- Single model family (DeepSeek-V4.1-Flash), small N (exploratory), no seed control (randomization-by-draws recorded as-is).
- The investigator is an experimenter channel, not a seat-based agent.
- Forced-assignment alliances measure compliance with notifications, not voluntary coordination.
- Suspicion disclosure is constrained by the available expression channels.
- No cross-game memory (audited harness behavior): repeated-play effects are unmeasured.
- **Dual-use**: the invitation template and harness quirks documented here are drawn from a public starter repository intended for safety research; we do not provide new attack techniques.

7. References

[1] Fish, S., Gonczarowski, Y. A., & Shorrer, R. I. (2024). Algorithmic Collusion by Large Language Models. arXiv:2404.00806.

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[3] Nakamura, M., Kumar, A., Das, S., Abdelnabi, S., Mahmud, S., Fioretto, F., Zilberstein, S., & Bagdasarian, E. (2026). Colosseum: Auditing Collusion in Cooperative Multi-Agent Systems. arXiv:2602.15198.

[4] Arike, R., Mehta Moreno, R., Subramani, S., Biswas, S., & Ward, F. R. (2026). How does information access affect LLM monitors' ability to detect sabotage? arXiv:2601.21112.

[5] Nguyen, K. L., Nghiem, H., & Tran, T. (2026). TraceGuard: Structured Multi-Dimensional Monitoring as a Collusion-Resistant Control Protocol. arXiv:2604.03968.

[6] Radev, N., Haas, L., Arnav, B., & Bernabeu-Pérez, P. (2026). The Best-Laid SCHEMEs: Coordinated Sabotage and Monitoring in Multi-Agent Systems. arXiv:2605.29178.

8. Reproducibility

The complete research package has been prepared as supplementary material (`our_experiment/`): RESEARCH_PLAN, PILOT_SPEC (protocol frozen at commit `d08fc09`), PROTOCOL_DEVIATIONS (D1-D8), EXPERIMENT_LOG, HARNESS_INTEGRITY_AUDIT, TRACK_3_2_ALIGNMENT, G3_REVIEW, SKEPTICAL_REVIEW, run manifests with SHA-256 hashes covering all 30 recorded runs (28 complete episodes; 2 partial runs flagged per the frozen INVALID rules), and the frozen analysis scripts (`analyze_g3.py`, `test_metrics.py` — 22 unit checks, `pilot_ledger.py`, `verify_numbers.py` — 32 audit checks recomputing every number in this paper from raw JSON). The package is archived at `github.com/itsuki2049/collusion-track32-study` (repository currently private — reviewers may request access from the authors at calpaliu@gmail.com; it will be made public upon the sprint's conclusion). Every reported figure regenerates via `python3 our_experiment/pilot_ledger.py` and `python3 our_experiment/verify_numbers.py`. If the submission portal does not accept supplementary files, the package will be released on a public repository upon the sprint's conclusion and is available from the authors in the interim (calpaliu@gmail.com).